

Palm Oil Suitability Layer (RI-2-9)

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Conservative binary suitability mask — research summary, thresholds & DEM recommendation
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Recommendation at a glance. DEM: Copernicus GLO-30 made **bare-earth** by subtracting the ETH Global Canopy Height (2020) model — both free, commercially-licensable, open datasets. Rule: **suitable = (slope $\leq 25^\circ$) AND (elevation $\leq 1,000$ m)**. Unsuitable = clearly too steep ($> 25^\circ \approx 47\%$) or too high ($> 1,000$ m a.s.l.). Terrain-only — no climate, soil, or economic filters — the deliberately generous “could it be terraced and planted” posture.

1. Purpose & framing

This summarises the research behind RI-2-9, the binary (suitable / unsuitable) palm oil suitability raster that feeds the EUDR RI (MH-49) and NDPE RI (MH-45). The design goal is conservative in the inclusive sense: if a hillside could reasonably be terraced and planted with oil palm, it is classified suitable. Only land that is unambiguously too steep or too high — even with terracing — is unsuitable. We deliberately exclude all financial / economic factors (price, demand, distance-to-mill, ROI) and set a physical threshold only.

“Conservative” therefore means we choose the generous upper-bound thresholds, not the yield-optimal ones. The mask answers only “could oil palm physically be planted here,” and is meant to be combined with land-cover, peat, protected-area and concession layers to become an actual risk product.

2. Slope threshold

Slope is the binding terrain constraint and the variable most directly tied to “can it be terraced and planted.” The literature gives a consistent ladder of management responses as slope increases; the conservative cutoff sits where the industry itself stops planting.

Slope (°)	Slope (%)	Management response / classification
0–6°	0–12%	Flat to undulating — plantable with no special measures (suitable).
~3–9°	~5–16%	Contour planting / soil-and-water conservation recommended (MPOB $\geq 3^\circ$; terracing of erodible soils above $\sim 6^\circ$).
~12–20°	24–38%	Hilly — marginal; contour terraces / platforms needed.
20–25°	38–47%	Somewhat steep — marginal to unsuitable; terracing mandatory. Top of the terraceable band.
$> 25^\circ$	$> 47\%$	Steep / very steep — not planted even with terracing. RSPO MY prohibits; MOSTA = unsuitable.

Recommended cutoff: unsuitable above 25° ($\approx 47\%$). This is (i) the explicit RSPO Malaysian regulatory line between plantable and prohibited¹; (ii) the top of the terraceable “marginal”

¹RSPO Malaysian National Interpretation (2015) and the RSPO “Oil Palm and Land Use Change” supplement (Harris et al. 2013), which defines steep terrain as average gradient $> 25^\circ$.

band in Malaysian soil-survey practice²; and (iii) the slope ceiling used in the key global study, Pirker et al. (2016)³, and the Kalimantan suitability study of Mantel et al. (2007)⁴. Indonesian GAP is slightly stricter at 40% $\approx$ 22°⁵. MPOB conservation guidance kicks in much earlier, from ~3–6°⁶. Because our definition of conservative is “include anything terraceable,” 25° is the right generous-but-defensible bound.

More-generous variant. If you want the absolute physical ceiling rather than the regulatory / agronomic one, use the Pirker / industry rule that slopes > 60% ($\approx$ 31°) should not be planted at all⁷. We recommend 25°/47% as primary and disclose 31°/60% as the sensitivity bound.

3. Elevation threshold

Elevation does not constrain oil palm directly — it is a proxy for temperature. The equatorial lapse rate is roughly –6 °C per 1,000 m, and growth slows below ~18 °C and effectively stops below ~15 °C.

Optimal / typical band: 0–500 m. Best growth at 0–500 m; the CIFOR / Austin refinement of Pirker capped Indonesian expansion at < 500 m.

Observed reality: very low. ~96% of 2010 SE Asian plantations sat below 200 m, and ~94% on land no steeper than 9%.

Conservative absolute limit: ~1,000–1,500 m. Pirker scores “suitable” up to 1,500 m; agronomic reviews cite ~1,000 m as the practical upper limit.

Recommended cutoff: unsuitable above 1,000 m a.s.l. This is conservative versus the 300–500 m optimal band and the $\leq$ 200 m operational reality, while staying below Pirker’s 1,500 m maximum⁸. A single flat 1,000 m cut is used at this stage; a latitude- or temperature-aware taper is intentionally out of scope and can be revisited later.

4. DEM recommendation & bare-earth approach

²Paramanathan, S. — Malaysian soil-survey suitability classes for oil palm on steep terrain, via MOSTA: hilly (24–38% / 12–20°) = marginal; somewhat steep (38–50% / 20–25°) = marginal-unsuitable; steep & very steep (>50% / >25°) = unsuitable.

³Pirker, J., Mosnier, A., Kraxner, F., Havlik, P. & Obersteiner, M. (2016). “What are the limits to oil palm expansion?” *Global Environmental Change* 40:73–81. Global, peer-reviewed, purely-biophysical model; slope class 0–25°, elevation 0–1,500 m (optimal 0–500 m); excludes economics.

⁴Mantel, S., Wösten, H. & Verhagen, J. (2007). “Biophysical Land Suitability for Oil Palm in Kalimantan, Indonesia.” Oil palm restricted to elevations < 500 m and slopes < 25°; dominant constraints are slope and poor drainage.

⁵Indonesian Ministry of Agriculture (2011), Good Agricultural Practices for oil palm — prohibits planting on slopes > 40% ($\approx$ 22°); cited in the EU REDD Facility oil-palm suitability protocol.

⁶MPOB, Oil Palm Bulletin 75 (Afandi et al.) — soil-and-water conservation recommended on slopes $\geq$ 3°; terracing of erodible soils above $\approx$ 6°; steepland planting causes severe erosion.

⁷EU REDD Facility (Pirker, 2017/2022), “Oil Palm Suitability Map” technical protocol (SW Cameroon). 30 m SRTM method: slope 0–12% ideal, plantable to 23%, >60% should not be planted at all; elevation 0–500 m fully suitable, scoring to 0 above 1,500 m; lapse rate –6 °C / 1,000 m.

⁸Pirker et al. (2016) score land “suitable” to an elevation ceiling of 1,500 m (optimal 0–500 m); our 1,000 m cut sits well inside this bound and is conservative everywhere in the belt.

The central problem: most free global DEMs are Digital Surface Models (DSM) that record the top of canopy / buildings, not bare ground — exactly where oil-palm deforestation risk is highest. A DSM can sit 20–40 m above true ground and distort slope at forest edges. A bare-earth Digital Terrain Model (DTM) is preferable for slope under canopy. The best bare-earth DTMs (FABDEM, FathomDEM) are released non-commercially, so they are unavailable to us; the only turnkey open bare-earth global model, MERIT, is 90 m — too coarse for slope.

DEM / layer	Res.	Type	Behaviour for slope in forest	Licence (open?)
Copernicus GLO-30	30 m	DSM	Modern gold-standard radar DEM; ~1.68 m global RMSE; partly canopy-penetrating but still a surface model.	Free, commercial OK (attribution).
ETH Canopy Height 2020	10 m	Canopy (input)	Not a DEM — subtract from GLO-30 to remove tree height and recover bare earth.	CC BY 4.0 (commercial OK).
SRTM 1" / 3"	30/90 m	DSM	Used by Pirker et al. 2016; 90 m too coarse for slope.	Free, commercial OK (public domain).
NASADEM	30 m	DSM	Reprocessed / improved SRTM.	Free, commercial OK.
ALOS AW3D30	30 m	DSM	Good optical-stereo quality.	Free commercial w/ attribution (JAXA).
ASTER GDEM	30 m	DSM	Noisier; artifacts.	Free, commercial OK.
MERIT DEM	~90 m	Bare-earth-corr.	Removes tree bias but 90 m too coarse for slope.	Free incl. commercial (conditions).

Recommendation: build a bare-earth surface from open inputs — GLO-30 minus ETH canopy height. Copernicus GLO-30⁹ is the best free, commercially-licensable 30 m global DSM. Subtracting the ETH Global Canopy Height (2020) layer¹⁰ — 10 m, CC BY 4.0, commercial use permitted — supplies the tree-height offset that GLO-30's radar only partly penetrates. This removes most of the 20–40 m over-height at forest edges that would otherwise inflate slope and shrink the suitable mask below its intended generosity. It reproduces, from fully-open inputs, what proprietary bare-earth DTMs do internally — with no licence cost and no non-commercial restriction. ETH canopy is a useful standalone layer for us to hold regardless.

5. Bare-earth correction (ETH canopy height)

⁹Copernicus DEM GLO-30 product documentation (ESA / Airbus). Global 30 m DSM from TanDEM-X; stated global vertical RMSE ≈ 1.68 m. Free & open licence permitting commercial use with attribution. Tiles on AWS Open Data (s3://copernicus-dem-30m/).

¹⁰Lang, N., Jetz, W., Schindler, K. & Wegner, J. D. (2023). "A high-resolution canopy height model of the Earth." *Nature Ecology & Evolution* 7:1778–1789. 10 m global canopy-top height from Sentinel-2 + GEDI (valid ~52°N–52°S). Released free of charge under CC BY 4.0 — commercial use permitted with attribution. <https://langnico.github.io/globalcanopyheight/>

The correction is a per-pixel subtraction on a common grid: bare-earth $\approx$ GLO-30 (DSM) – canopy-height. In practice a few rules keep it well-behaved:

Grid alignment. Resample the 10 m ETH canopy to the exact GLO-30 30 m grid using average resampling, so each 30 m cell gets the mean canopy height over its footprint (correct for a height-offset subtraction).

No-canopy = 0. Non-forest, water and gaps carry no canopy; set canopy nodata to 0 so those pixels pass through as pure DSM.

Clamp over-subtraction. Cap canopy at the DSM value (subtract $\min(\text{canopy}, \text{DSM})$) so noisy canopy pixels can never drive bare-earth below plausible ground.

Smooth lightly. The canopy model carries its own error, so the subtracted surface has spurious micro-relief; a light 3×3 low-pass before slope suppresses it without flattening real terrain.

GEDI-derived canopy (and therefore the ETH model) is valid to roughly 52°N – 52°S — comfortably covering the entire oil-palm belt. This is a modelled subtraction, not a physics-based ground retrieval, so it should be validated against reference lidar or a known-terrain pilot before the steep-edge pixels are trusted (see Caveats).

6. Slope calculation method

Algorithm: compute slope with Horn’s method (3×3 window)¹¹ — the GDAL / ArcGIS default, robust to radar-DEM noise.

Resolution trap: slope decreases as DEM resolution coarsens; 90 m systematically under-estimates steep slopes versus 30 m¹². For a conservative mask that must not under-call steepness, use 30 m and do not resample 90 m up.

Units: threshold in degrees (clean 25° rule); report the percent equivalent ($25^\circ = 46.6\% \approx 47\%$).

Projection gotcha: on a geographic (EPSG:4326) DEM, `gdaldem` needs a scale (`-s 111120`) to convert degrees to metres, and that single scale is only exact at the equator — E–W pixel spacing shrinks with $\cos(\text{latitude})$, ~13% smaller at 30° . Across the tropical belt this biases slope; for best accuracy reproject tiles to a metric CRS (per-UTM zone) or apply a per-row $\cos(\text{lat})$ correction (trivial in the Python path below).

7. Processing methodology (global tropical belt)

Target coverage is the full pan-tropical oil-palm belt — roughly 30°N to 30°S — spanning SE Asia, the Congo Basin and Central/South America. Both inputs are open and available as tiles: GLO-30 as $1^\circ \times 1^\circ$ COGs on AWS Open Data, ETH canopy as $3^\circ \times 3^\circ$ GeoTIFFs. The pipeline is:

¹¹Slope computed with Horn’s method (3×3 window) — the default in GDAL “`gdaldem slope`” and ArcGIS; more robust to DEM noise than Zevenbergen–Thorne for radar DEMs.

¹²DEM-resolution effect on slope: coarser grids smooth peaks/valleys and systematically under-estimate steep slopes (USLE/RUSLE LS-factor literature).

fetch → mosaic (VRT) → align canopy to DEM grid → subtract → slope → threshold → cloud-optimize.

Option A — GDAL command-line. A pure-GDAL bash pipeline is fully reproducible and needs no code. VRTs keep the mosaics virtual; gdalwarp is multi-threaded; the whole chain is auditable step by step.

```
# ---- 0. AOI: pan-tropical oil-palm belt, 30 N to 30 S ----
# -te uses xmin ymin xmax ymax in EPSG:4326
export TE="-180 -30 180 30"
export RES=0.0002777778 # 1 arc-second (~30 m) in degrees

# ---- 1. Fetch tiles intersecting the belt (AWS Open Data, no auth) ----
# Copernicus GLO-30 : s3://copernicus-dem-30m/ (1x1 deg COG tiles)
# ETH canopy 2020 : 3x3 deg GeoTIFF tiles (ETH bucket / GEE export)
aws s3 cp --no-sign-request --recursive s3://copernicus-dem-30m/ ./glo30/ \
  --exclude "*" --include "Copernicus_DSM_COG_10_*" # keep only belt tiles

# ---- 2. Virtual mosaics (no monster files on disk) ----
gdalbuildvrt glo30.vrt glo30/*/*_DEM.tif
gdalbuildvrt eth_ch.vrt eth/*.tif

# ---- 3. Put canopy on the exact GLO-30 grid (10 m -> 30 m, mean height) ----
gdalwarp -t_srs EPSG:4326 -te $TE -tr $RES $RES -tap \
  -r average -dstnodata 0 \
  eth_ch.vrt eth_on_glo30.tif -multi -wo NUM_THREADS=ALL_CPUS

# ---- 4. Bare earth = DSM - canopy (clamp so canopy can't exceed DSM) ----
gdal_calc.py -A glo30.vrt -B eth_on_glo30.tif \
  --calc="A-minimum(B,A)" --outfile=bare.tif --type=Float32 --co COMPRESS=DEFLATE

# ---- 5. Slope in degrees, Horn 3x3; -s scales deg->m (equatorial) ----
gdaldem slope bare.tif slope_deg.tif -alg Horn -s 111120 -compute_edges

# ---- 6. Binary suitability: slope <= 25 AND elevation <= 1000 ----
gdal_calc.py -A slope_deg.tif -B bare.tif \
  --calc="(A<=25)*(B<=1000)" --outfile=suitable.tif --type=Byte

# ---- 7. Cloud-optimize + build overviews ----
gdal_translate suitable.tif suitable_cog.tif -of COG -co COMPRESS=DEFLATE
gdaladdo -r nearest suitable_cog.tif 2 4 8 16
```

Pros: zero custom code, battle-tested, transparent, trivially re-runnable, streams via VRT.

Cons: gdal_calc.py reads whole bands into memory, so at belt scale the subtract/threshold steps will exhaust RAM unless you tile the AOI (loop over 5°×5° blocks, then gdalbuildvrt the outputs); large intermediates (bare.tif, slope_deg.tif) are written to disk between steps; and the single -s slope scale carries the cos(lat) bias noted above.

Option B — Python (rioxarray + dask). For the full belt in one pass, a chunked Dask graph is materially better: it processes out-of-core (never loading the whole raster), fuses subtract + slope + threshold so no big intermediates hit disk, parallelises across cores/tiles, and lets you apply the correct per-row cos(lat) metric slope.

```
import rioxarray as rxr, xarray as xr, numpy as np

# lazy, chunked reads -> nothing loads into RAM until .compute().write
dsm = rxr.open_rasterio("glo30.vrt", chunks={"x":4096,"y":4096}).squeeze()
ch = rxr.open_rasterio("eth_on_glo30.tif", chunks={"x":4096,"y":4096}).squeeze()

# bare earth: clamp canopy to DSM, treat no-canopy as 0
bare = dsm - xr.where(ch > dsm, dsm, ch).fillna(0)
```

```
# per-row cos(lat) correction so E-W spacing stays metric across the belt
res = 0.0002777778
dx = 111320 * np.cos(np.deg2rad(bare["y"])) * res      # m / pixel, varies by row
dy = 110540 * res

# gradient with dask.map_overlap so 3x3 windows are correct at chunk edges
def horn_slope(a):
    gy, gx = np.gradient(a)
    return np.degrees(np.arctan(np.hypot(gx / dx.values[:,None], gy / dy)))
slope = xr.apply_ufunc(horn_slope, bare, dask="parallelized",
                      output_dtypes=[np.float32])

suitable = ((slope <= 25) & (bare <= 1000)).astype("uint8")
suitable.rio.to_raster("suitable_cog.tif", driver="COG",
                    compress="DEFLATE", windowed=True)
```

Pros: out-of-core and parallel (handles the whole belt without OOM), one fused pass (no intermediate GeoTIFFs), exact per-latitude slope, and full control over clamping/nodata logic. Use `dask.map_overlap` (or `richdem`) so 3×3 slope windows are correct at chunk edges. **Cons:** more code to write and maintain, chunk-size and CRS handling need care, and reproducibility depends on pinning the environment.

Recommendation: prototype and validate on one country with the GDAL CLI (fast, transparent, easy to sanity-check), then run the full tropical belt with the Python/dask path — or, if staying in bash, tile the GDAL pipeline into 5° blocks and mosaic. Either way, reproject to a metric CRS (or apply the cos-lat correction) before deriving slope so steepness is not latitude-biased.

8. Implementation plan

Stage 1 (ship now): GLO-30 – ETH canopy → bare-earth → Horn slope (degrees) → binary rule `suitable = (slope ≤ 25°) AND (elevation ≤ 1,000 m)`, output as a cloud-optimised GeoTIFF over the 30°N–30°S belt. Cite Pirker 2016, RSPO MY-NI, Mantel 2007, Copernicus GLO-30, and Lang et al. 2023 (ETH canopy). Elevation may use raw GLO-30 or bare-earth interchangeably — the canopy offset (<40 m) is negligible against the 1,000 m cut.

9. Caveats

“Conservative” cuts two ways — state which in metadata. Our definition maximises suitable area. 25°/1,000 m is the regulatory / agronomic ceiling; 31°/1,500 m is the only more-generous defensible (absolute-physical) variant.

Slope is area-averaging-dependent. RSPO’s 25° is a mean over a 25 ha block; a per-pixel 30 m slope flags more, finer steep pixels — making per-pixel thresholding slightly stricter (less generous) than the RSPO block rule. A focal-mean window before thresholding reproduces the regulatory intent more faithfully.

Bare-earth correction is approximate. Subtracting a modelled canopy height is not a physics-based ground retrieval; residual canopy-model error (and GEDI’s few-metre height uncertainty) propagates into slope. Validate against reference lidar / GEDI ground returns or a

known-terrain pilot before trusting steep-edge pixels, and smooth lightly to suppress subtraction noise.

Economics deliberately excluded. Real siting is dominated by distance-to-mill and roads; the mask will flag terrain that would never be developed economically — correct and intended for a physical-possibility mask, but worth stating to users.

Sources

Full citations appear as numbered footnotes throughout. Key references: Pirker et al. 2016 (Global Environmental Change); RSPO / Harris et al. 2013; Paramananthan / MOSTA; Indonesian MoA GAP 2011; EU REDD Facility (Pirker) protocol; Mantel et al. 2007; MPOB OPB 75; Copernicus GLO-30 documentation; Lang et al. 2023 (ETH Global Canopy Height, CC BY 4.0).